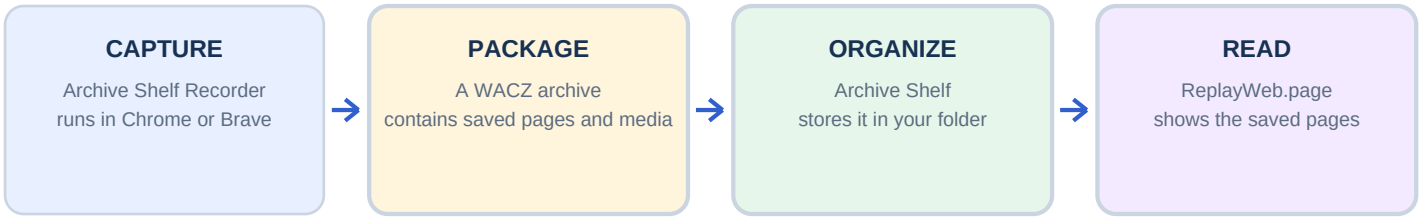
ARCHIVE SHELF

A plain-language guide to what it does, what runs, and how to read its source code

In a sentence: the Recorder saves pages as you browse, Archive Shelf files the WACZ, and ReplayWeb.page opens it for you.



What changed in this version

The capture engine is now bundled into the Archive Shelf Recorder extension. One extension can ask for a Shelf folder, start recording, and save the finished WACZ directly into that folder. You do not need a separate ArchiveWeb.page recorder extension or the ReplayWeb.page desktop app.

The ReplayWeb.page reader files are bundled inside the app, so opening a saved archive does not download viewer code from a website. The reader opens at replay.localhost, a separate local browser address that keeps archived page scripts away from the organizer's folder controls.

App folder: the folder containing Start Archive Shelf.bat

1. What you click

Think of it like a camera and a filing cabinet. The Recorder watches the browser while you visit pages. Archive Shelf is the cabinet where the finished archive is filed.

Step	What you do	What happens
1	Start Archive Shelf	Double-click Start Archive Shelf.bat. Keep its black server window open. The browser page is the screen; Python runs in the background.
2	Choose a Shelf folder	Click Archive Shelf Recorder in Chrome or Brave. Choose a folder by clicking through the folder list, then choose Save in this folder.
3	Start archiving	Click Start Archiving. The current page reloads so recording can begin before its content loads.
4	Browse normally	Click links and visit the pages you want to keep. Let images, video, and other content load while recording is on.
5	Stop and save	Open the Recorder button again and click Stop and Save WACZ. Wait for the saved message. The archive appears in the chosen Shelf folder.

If you close the little extension popup, recording continues. Open it again when you are ready to stop and save.

What is a WACZ?

A WACZ is a single archive package. It can contain saved web pages, pictures, scripts, other loaded resources, and an index that helps the reader find them. Archive Shelf stores a copy of that package as one file.

This is not a whole-site download button. Only pages you visit and resources that load during recording are captured. To include a link, open it while recording is on and wait for its content to load.

2. What is happening behind the button

Here is the simple version of the machinery inside the browser:

A. The extension starts a session

The Shelf Recorder makes a new recording collection in browser storage and connects to the active browser tab.

B. The browser reports traffic

The recorder uses Chromium's debugging interface to observe page requests and responses as the site loads. This is why Chrome or Brave asks for the debugging permission and may show a recording banner. Cookies are turned off by default to reduce the chance of saving login information.

C. You browse

As you open pages, the recorder saves the page and the media/resources that the browser actually loads. Opening a link in another tab can also be recorded when the browser connects it to the active recording.

D. Stop makes the package

When you stop, the Recorder's WACZ writer gathers the saved recording data and makes a WACZ stream.

E. Python files it

The extension sends that stream to the Archive Shelf server on this computer. Python saves the file under library and adds its name and folder to archive-shelf-data.json.

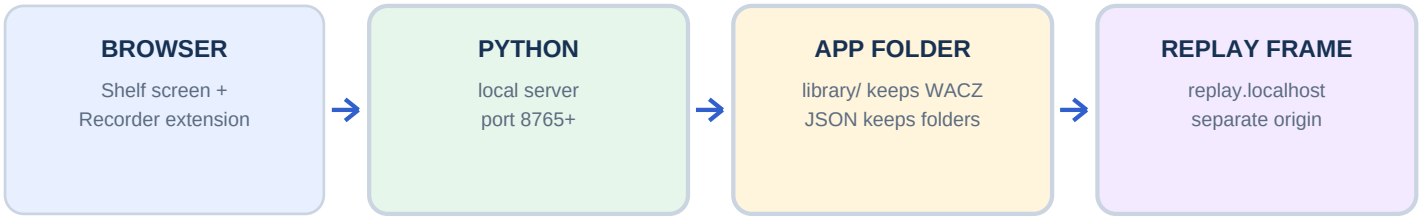
The new extension replaces both jobs: it captures the pages and sends the finished archive to Archive Shelf. The old importer is no longer needed for new recordings.

Limits to know

- It saves pages and media that load while recording. It does not crawl every unvisited link on a website.
- Some sites limit capture, and browser-internal pages such as chrome:// pages cannot be recorded like ordinary websites.
- If the browser closes before Stop and Save finishes, the final WACZ may not have been created or filed yet.
- The recorder needs the Chrome or Brave debugging permission to observe page traffic.

3. The local address, files, and internet

localhost and 127.0.0.1 both point to this computer. The number after the colon is a local port, like a numbered door. The launcher tries port 8765 first, then a nearby port if it is busy. The extension connects through 127.0.0.1; the organizer screen opens at localhost.



Connection	What it means
Recorder to Shelf	The extension sends the WACZ to the local Python server on this same computer.
Shelf to archive	The replay frame reads the selected WACZ file from library/ on this computer.
Shelf to viewer code	ReplayWeb.page's UI and service worker are local files in replay/. You do not need internet to open an archive.
Replay isolation	Archived pages run under replay.localhost. The server allows that address to read archive files, but it does not provide the organizer API there.
Recorder to website	The browser still visits the website you are archiving. The extension does not upload the finished WACZ to a cloud service.

Windows, Linux, and another computer

On Windows, both .bat launchers find Python 3.13 at %LOCALAPPDATA%\Programs\Python\Python313\python.exe. Windows expands %LOCALAPPDATA% for the signed-in account, so no username is stored in the launcher. If Python is installed elsewhere or another version is used, edit the PYTHON_EXE line in both launchers. See Archive Shelf Python Path Guide.pdf in the Version 1.0.0 folder for step-by-step instructions.

The organizer server is Python and can run on Windows or Linux. The .bat launchers are for Windows; on Linux, start organizer_server.py with Python 3. The browser extension needs Chrome or Brave on the computer where you capture. To move everything, copy the app folder together with library/ and archive-shelf-data.json. On another computer, install Python and Chrome or Brave, start Archive Shelf, and load the Recorder extension there.

The local server has no sign-in screen and archive files are not encrypted. Keep the app on a trusted computer account. Do not expose its port to Wi-Fi or the internet.

4. Your folders and archive files

Archive Shelf is a file organizer, not just a list of recent recordings. Folders can be nested, for example: MTG / JumpStart / Avatar the Last Airbender.

Ways to organize

- Create a folder at the top level or inside the folder you are viewing.

- Choose archive files or drag WACZ files from Windows Explorer into the Shelf page.

- Move an archive by dragging its row onto a folder. To move it up, drop it on Parent folder. To move it to the root, drop it on All folders.

- Click an archive name to open it inside Archive Shelf. Remove deletes Archive Shelf's copy; it does not change your original file.

The two important data parts

Item	Think of it as	Keep it when...
archive-shelf-data.json	The card catalog: names, folders, and IDs.	You move or back up your Shelf collection.
library/	The cabinet: the actual WACZ archive copies.	You want to keep the saved websites.
browser-extension/Archive Shelf Recorder/	The ready-to-load browser extension.	You install the recorder in Chrome or Brave or move it to another computer.
browser-extension/archiveweb-page-source/	Readable source used to build the extension.	You inspect or upgrade the recorder code.

When backing up your collection, keep library/ and archive-shelf-data.json together. The source code is useful for rebuilding the program, but it does not contain your archived pages.

5. How to inspect the code

The app is not hidden inside the chat. The working files are on your computer. Open the .py, .html, .js, .ts, .json, .svg, and .bat files in Notepad or a code editor such as VS Code.

File or folder	What it controls	Look here when...
index.html	Shelf screen, folders, drag and drop, archive viewer.	You want to change the screen or what the buttons do.
organizer_server.py	Local server, imports, moving files, folders, saved catalog.	You want to change storage or server behavior.
replay-viewer.html and replay-viewer.js	Separate ReplayWeb.page frame and selected archive URL.	You want to change how archived sites open inside the organizer.
archiveweb-page-source/static/shelf-popup.html and shelf-popup.js	Recorder popup screen and folder picker.	You want to change the extension's look or steps.
archiveweb-page-source/src/ext/bg.ts	Extension background events and sending WACZ to Archive Shelf.	You want to change how recording starts, stops, or saves.
archiveweb-page-source/src/ext/browser-recorder.ts and src/recorder.ts	Upstream Webrecorder capture logic.	You are studying how browser traffic is recorded.
archiveweb-page-source/src/sw/downloader.ts	Upstream WACZ/WARC file writer.	You are studying how recording data becomes an archive file.
check_replaywebpage_update.py	Checks and updates local ReplayWeb.page package files.	You are changing the reader update process.

Two ways to read the code

- Open Archive Shelf Source Code.txt for one copyable document containing the main app and recorder files.

- Open browser-extension/archiveweb-page-source for the complete readable upstream project plus the Archive Shelf changes. Its LICENSE.md explains the upstream license.

- The ready-to-load extension folder contains built files. For understanding or editing, use the readable source project, not the large compressed bg.js file.

6. Updates, permissions, and getting help

Updates are split into two different things:

Updater or code	What it updates
Check ReplayWeb.page Updates.bat	Checks npm for a newer ReplayWeb.page package. If you approve, it verifies SHA-512, backs up the reader files, and updates the local UI and service worker.
Archive Shelf Recorder source	The capture extension. The reader updater does not change it. Rebuilding this extension needs an updated source project and a new build.
Archive Shelf Python and screen	These are the organizer app files. Updating the reader does not change them.

The server checks local Host and Origin headers, limits archive uploads, and keeps the organizer API on localhost/127.0.0.1. Archived page scripts use replay.localhost, which cannot access those organizer routes. The recorder is adapted from the open-source Webrecorder ArchiveWeb.page project. Its source is kept with this app, and its LICENSE.md is included.

Quick troubleshooting

What you see	What to try
Recorder says Shelf is not running	Start Archive Shelf and leave the black server window open. Click Refresh folders.
No folder appears	Create it in Archive Shelf, then reopen the folder chooser or press Refresh folders.
Archive does not appear after stopping	Wait for the Saved message. Keep the app server running, then refresh the Shelf page.
A page or media is missing	Record again, visit that page while recording, wait for the content to load, then stop and save.

Before changing code: stop the app, copy the whole app folder as a backup, and keep library/ plus archive-shelf-data.json. A different project in a different folder will not overwrite this collection.

References: https://github.com/webrecorder/archiveweb.page | https://archiveweb.page/en/usage/ | https://replayweb.page/docs/embedding/